

Run 01 | Business exit

Global (including India)

IV only

Target Entrepreneurial/business exit in the past 12 months; binary 1 vs. 0
Sample 1,331,428 respondents **Coverage** 89 countries; 2015–2022
Positive cases 61,654 (4.63%) **Case time** 30m 05s
Train / val. / test 905,374 / 159,770 / 266,284 (68% / 12% / 20%, stratified)
SCT predictors (12): Self-efficacy, Opportunity perception, Fear of failure, Knowing entrepreneurs, Entrepreneurial intention, Media coverage, Business angel, Established business owner, Good career choice, Ease of starting a business, High status, Social enterprise.
Legatum predictors (13): Prosperity index score, Safety and security, Personal freedom, Governance, Social capital, Investment environment, Enterprise conditions, Market access and infrastructure, Economic quality, Living conditions, Health, Education, Natural environment. Total independent variables: 25.
Controls (0): None. Country and respondent ID are excluded as predictors.
Protocol: Ten CPU models; model seed 20260925; fixed splits by sample/outcome; training-only preprocessing; F1 threshold selected on validation data. Test scores below are unweighted.

Result in brief

Highest observed ROC-AUC: XGBoost (0.7874). Highest F1: XGBoost (0.2413). The all-negative accuracy baseline is 95.37% (full sample).

Model performance on held-out test records

Model	ROC-AUC	PR-AUC	Acc.%	Prec.%	Recall%	F1	Time
XGBoost	0.7874	0.1919	89.56	18.18	35.86	0.2413	23s
CatBoost	0.7837	0.1885	90.29	18.77	32.95	0.2391	1m 47s
H2O GBM	0.7795	0.1793	90.27	18.52	32.41	0.2357	1m 24s
Random Forest	0.7790	0.1950	91.12	19.95	30.42	0.2409	4m 08s
Lasso	0.7612	0.1407	88.27	15.19	33.46	0.2089	4m 50s
Elastic Net	0.7612	0.1407	88.27	15.18	33.45	0.2089	5m 11s
Logistic Regression	0.7611	0.1406	88.29	15.21	33.43	0.2090	58s
Ridge	0.7605	0.1401	87.50	14.85	35.94	0.2102	7m 03s
Naive Bayes	0.7036	0.1007	87.67	12.00	26.25	0.1647	3m 21s
Decision Tree	0.5000	0.0463	4.63	4.63	100.00	0.0885	28s

Models ranked by ROC-AUC; small differences are descriptive, not significance tests. Time includes fit, prediction and output processing. Logistic regression uses a small stabilizing penalty; Lasso/Ridge/Elastic Net use 3-fold training CV.

Random-forest feature importance: top 8

Rank	Predictor	Raw importance	Share%
1	Entrepreneurial intention	1,587.92	9.37
2	Business angel	1,109.63	6.55
3	Good career choice	687.13	4.06
4	Fear of failure	653.34	3.86
5	Self-efficacy	621.18	3.67
6	Education	566.64	3.34
7	Opportunity perception	532.42	3.14
8	Media coverage	527.61	3.11

Share is the percentage of total forest impurity importance across all 49 encoded predictors, not explained variance. Raw values are not comparable across samples. Missing indicators are derived preprocessing features.

Leading Legatum predictors within the same forest

Rank	Legatum predictor	Raw importance	Share%
1	Education	566.64	3.34
2	Prosperity index score	504.18	2.98
3	Living conditions	488.87	2.89

Legatum subset ranked by raw importance; shares use total forest importance, not the Legatum subtotal.

Source: Completed Objective 5 batch, 29 September 2026; 01_obj5_global_iv_EXIT_ENT_harmonized. Scores and importance are taken directly from saved CSV outputs.